

• market_signals_v2_2026_05

Signal-Tune v2

Gap Analysis

A structured comparison of the current pipeline implementation against the LaunchHouse v2 market-signals scoring model — identifying what exists, what is missing, and what must change to reach acceptance criteria.

PREPARED
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AUDIENCE
Engineering Leadership

SOURCE OF TRUTH
LaunchHouse Signal-Tune v2
PDF + SUMMARY.md

Three agents. Five gaps. One scoring model that needs a full replacement.

The current four-agent pipeline (Research → Enrichment → Scoring → Personalization) is structurally sound but calibrated to an outdated 6-signal, firmographic-heavy scoring model. The v2 model introduced by LaunchHouse inverts the weight distribution — moving mass from static company attributes toward dynamic operational signals. The majority of required changes are **prompt and config only**; no schema migrations are needed.

CORE STRATEGIC SHIFT

"Your agent should not ask: 'Who uses Cvent?' It should ask: 'Which Cvent-using event team has a real operating reason to need execution help now?' Cvent usage is a gate, not a signal. Scores should reflect compound evidence of operational pressure."

CRITICAL · FIX NOW

Scoring Model — Full Replacement

6-signal prompt must be replaced with the v2 8-signal model including 7 hard caps and a compound-evidence Hot gate.

CRITICAL · FIX NOW

4 Research Query Sets Missing

Hiring (ATS-targeted), Cvent complexity, MarTech pressure, and Cvent Certified are not queried at all today.

CRITICAL · FIX NOW

Enrichment Output Schema Incomplete

5 structured signal blocks required by v2 are absent from EnrichmentAgent output: hiring signals, complexity, certified profile, capacity stress index, source confidence.

MEDIUM · PHASE 4

PersonalizationAgent Routing

No structured routing table exists.

`recommended_template_driver` and 4 hook fields are not written to `personalization_hooks`.

The scoring model needs to be replaced, not tuned.

The current `SCORING_SYSTEM_PROMPT` defines 6 signals summing to 1.0 weight. The v2 model defines 8 signals, rebalances every weight, introduces 7 hard score caps, and adds a compound-evidence gate that prevents a single weak signal from producing a Hot lead.

Signal Weight Redistribution



Score Caps — None Currently Implemented

The current agent has no cap logic. All 7 caps below are absent, meaning a lead with only Cvent confirmation can score Hot today.

Scenario	v2 Cap	Current Behaviour
Only proof is that the company uses Cvent	45 max	● No cap — can score Hot
No verified event, no hiring signal, no complexity	65 max	● No cap
No relevant buyer / persona found	70 max	● No cap

Scenario	v2 Cap	Current Behaviour
Event inferred but no specific future date	Temporal capped at 40% of urgency weight	• No cap
Low source confidence across key claims	Cannot be Hot	• No source-confidence gate
Job posting with no Cvent / platform language	Capacity signal capped at Medium	• No cap
Generic funding news only	No lift unless tied to event expansion evidence	• No cap

Additionally: Hot tier now requires ≥ 2 strong dynamic signals — a single high signal cannot produce Hot.

SECTION 2 — RESEARCH AGENT

Four of six required query sets are missing.

The current `ResearchAgent` runs 5 generic queries (news, hiring, tech stack, domain, contact LinkedIn). The v2 model defines 6 targeted query sets with specific source priority order, keyword libraries, and evidence extraction targets.

Query Set	Current State	v2 Requirement	Status
Company news	<code>"{company} company news recent"</code>	Same intent	● Aligned
Cvent event pages	SerpAPI <code>site:cvent.com</code> search	Same intent	● Aligned
Verified event urgency	No dedicated query	Official event pages, Cvent registration URLs, confirmed future dates. Reject snippets with no date.	● Missing
Hiring & capacity gap	<code>"{company} event marketing team hiring"</code> — generic, no ATS targeting	ATS-targeted queries: Greenhouse → Lever → Workday → Ashby → SmartRecruiters → iCIMS → Jobvite. Extract Cvent terms in JD, posting age.	● Partial
Cvent complexity signals	Not queried	Module names (Attendee Hub, OnArrival, Appointments, housing, Data Bridge), integration targets, complexity descriptors.	● Missing
MarTech & reporting pressure	Not queried	Salesforce, HubSpot, Marketo, Snowflake, Data Bridge, attribution, CRM sync, event ROI language.	● Missing
Cvent Certified profile	Not queried	<code>"Cvent Certified"</code> OR <code>"#CventCertified"</code> per contact + compound rule: certified person left + active hiring → <code>capacity_stress_index: high</code> .	● Missing

SECTION 3 — ENRICHMENT AGENT

Five structured signal blocks are absent from the output schema.

The v2 model requires `EnrichmentAgent` to extract and store structured signal blocks into the existing `enrichment_data.data` JSONB column. No migrations are needed — only optional fields added to the Pydantic output model and updated prompt instructions.

Output Field	Current	v2 Requirement	Status
<code>company</code> <code>firmographics</code>	Size, revenue, industry, tech stack	Same	● Aligned
<code>contact</code> <code>seniority</code>	Seniority, department, buyer_persona	Same intent	● Aligned
<code>event_hiring_signals</code>	Not extracted	<code>has_open_cvent_role</code> , <code>role_titles_found</code> , <code>cvent_terms_in_jd</code> , <code>posting_age_days</code> , <code>signal_strength</code>	● Missing
<code>cvent_complexity_signals</code>	Not extracted	<code>modules_detected</code> , <code>integration_targets</code> , <code>complexity_level</code> , <code>source_urls</code>	● Missing
<code>cvent_certified_signal</code>	Not extracted	<code>found</code> , <code>person_relevance</code> , <code>certification_phrase</code> , <code>confidence</code>	● Missing
<code>program_pattern_signals</code>	Loosely in <code>event_program</code> dict — unstructured	Typed array: <code>program_type</code> , <code>matched_title</code> , <code>repeatability</code> , <code>confidence</code>	● Unstructured
<code>capacity_stress_index</code>	Not extracted	<code>"high"</code> / <code>"medium"</code> / <code>"low"</code> compound signal (certified person left + active hiring = high)	● Missing
<code>source_confidence_summary</code>	Not extracted	Per-evidence confidence tier (official page > ATS > press release > snippet > inferred)	● Missing
<code>recommended_template_driver</code>	Not extracted	Primary routing key consumed by PersonalizationAgent	● Missing

SECTION 4 — PERSONALIZATION AGENT

No routing table. No hook fields. No template driver.

The current PersonalizationAgent has no structured routing logic tied to the primary signal driver. The v2 model defines 7 explicit routes, 5 output fields, and explicit "avoid" rules per template.

Primary Signal Driver	Template	Messaging Angle	Currently Wired
Verified event 31–120 days out	T8 Build Scoping	Scope is being locked; fixed-fee Cvent implementation before the rush	• No
Verified event 0–30 days out	T7 Rush	Final-stretch support for registration, Attendee Hub, integrations	• No
Event / Cvent hiring gap	T6 or capacity-led T1	Bridge Cvent/event ops capacity while internal team scales	• No
Cvent complexity	T1/T6 or T8 if event exists	Registration architecture, multi-track, onsite/mobile, integrations	• No
MarTech / reporting pressure	T6 or T9 if news-triggered	Cleaner Cvent setup improves downstream reporting and CRM data flow	• No
Program pattern only	T6 Fit-Based	Overflow capacity for repeatable event programs	• No
Cvent Certified only	No dedicated template	Background signal only — never use as cold outreach hook	• No

SECTION 5 — IMPLEMENTATION PLAN

Five phases. Prompt-and-config only. No migrations.

All changes fit within the non-negotiable constraint from the v2 spec: prefer prompt/config changes and existing JSONB storage. No new database tables or data providers.

Phase	Change	Code Impact	Acceptance Gate
1	Replace scoring prompt with v2 8-signal model + 7 caps	ScoringAgent prompt/config only	QA-01 through QA-06 pass; no single-signal Hot leads
2	Add 4 targeted query sets to ResearchAgent	ResearchAgent query list + prompt. No new data provider.	Hiring, complexity, MarTech, Certified queries return evidence for known accounts
3	Extract structured signal blocks in EnrichmentAgent	Optional Pydantic fields in EnrichmentOutput + enrichment prompt update	5 new JSONB blocks present in enrichment_data for v2-enriched leads

Phase	Change	Code Impact	Acceptance Gate
4	Add score cap logic to ScoringAgent	<code>ScoringAgent</code> prompt + optional post-processing in <code>scoring_agent.py</code>	Cvent-only leads never exceed 45; no-buyer leads never exceed 70
5	PersonalizationAgent routing + hook fields	Routing decision logic + <code>personalization_hooks</code> fields in output	<code>recommended_template_driver</code> present for all scored leads

SECTION 6 — QA ACCEPTANCE SCENARIOS

6 scenarios define done.

QA-01 · RED-HOT COMPOUND

Verified event 58 days out + open Event Ops Manager (Cvent, Attendee Hub, Salesforce) + Director of Events contact

Hot · 85-95

Route → T8 with capacity-aware personalization

QA-02 · HOT EVENT COMPLEXITY

Annual summit 80 days out + multi-track sessions + Head of Field Marketing

Hot · 78-88

Route → T8 build scoping angle

QA-03 · CAPACITY-LED WARM

Open Cvent Administrator + Marketing Ops role; no verified event

Warm · 65-75

Route → T6 or capacity-led T1; re-enrich monthly

QA-04 · PROGRAM PATTERN

Annual incentive trip / Presidents Club with registration/travel details; no Cvent evidence

Warm · 58-72

Route → Fit-based sequence; no false Cvent claim

QA-05 · CERTIFICATION ONLY

Cvent Certified profile, no event timing / hiring / complexity / reporting signals

Warm max

No certification-led email hook

QA-06 · NURTURE

Known Cvent user in high-fit industry; no event timing, hiring, complexity, or pressure

Cold · 35-50

Hold or low-touch

HARD ACCEPTANCE RULES

Implementation is complete only when: (1) all 6 QA scenarios hit expected tier and routing; (2) no lead scores Hot on a single weak signal; (3) Cvent usage alone never produces a Hot lead; (4) every Hot-score lead includes a high-confidence source URL for its primary signal driver; (5)

`recommended_template_driver` is present in `message.personalization_hooks` for all scored leads.